COSMOLOGY

Problem Set 5: Perturbation Theory

Hieu Nguyen

These are the end of chapter problems for Chapter 5 of the notes. Equation numbers refer to the full notes.

1. A Polytropic Fluid

Consider a universe filled with a non-relativistic fluid with $P = \alpha\rho^{4/3}$, where $P \ll \rho$, so that the background evolution is well-approximated as matter dominated. Inside the Hubble radius, a Fourier mode of the density contrast $\delta \equiv \delta\rho/\rho$ evolves as

$$\delta'' + \mathcal{H}\delta' + \left(c_s^2 k^2 - \frac{3}{2}\mathcal{H}^2\right)\delta = 0,$$

where $c_s^2 \equiv \partial P/\partial\rho$, primes denote derivatives with respect to conformal time η and $\mathcal{H} \equiv a'/a$.

a). Show that $c_s^2 = c_0^2\left(\eta_0/\eta\right)^2$, where c_0 and η_0 are constants.

b). Describe qualitatively how you expect the solution for the density contrast to behave on small and large scales.

c). Find power law solutions of the form $\delta \propto \eta^\beta$, where β depends on the wavenumber k. At what value of k does the solution change from monotonic to oscillatory? For what k is the growing mode a constant?

2. The Jeans Mass Of Baryons

a). Before recombination, the baryons are tightly coupled to photons, and the sound speed is $c_s \approx c/\sqrt{3}$. Show that the Jeans length is then comparable to the Hubble radius, so baryon perturbations can't grow inside the horizon.

b). After recombination, the baryons are a gas of hydrogen at temperature $T \approx T_{\rm CMB}$ (about 3000 K at $z = 1100$). Compute the Jeans mass $M_J = \frac{4\pi}{3}\bar{\rho}_m\left(\frac{\lambda_J}{2}\right)^3$ and compare it to the mass of a globular cluster ($\sim 10^5$–$10^6 M_\odot$).

3. Velocities From Densities

Using the linearized continuity equation, show that in linear theory the peculiar velocity field is

$$\boldsymbol{v}_{\boldsymbol{k}} = i\frac{aHf}{k^2}\boldsymbol{k}\,\delta_{\boldsymbol{k}},$$

where $f = d\ln D_+/d\ln a$. What is the typical peculiar velocity of a galaxy today on a scale where $\delta \sim 1$ and $k \sim 0.1\,h\,\mathrm{Mpc}^{-1}$?

4. Growth With Massive Neutrinos

Suppose a fraction f_ν of the matter is in neutrinos, which free-stream and don't cluster on small scales. On those scales, only $1 - f_\nu$ of the matter contributes to the source term, $\ddot{\delta} + 2H\dot{\delta} = 4\pi G\bar{\rho}_m(1 - f_\nu)\delta$. Show that in matter domination $\delta \propto a^p$ with $p \approx 1 - \frac{3}{5}f_\nu$. Over the roughly $\ln(a_0/a_{\rm eq}) \approx 8$ e-folds since equality, by how much is the small-scale power spectrum suppressed?

5. Spherical Collapse

Using the cycloid solution, show that the density contrast of the spherical region is

$$1 + \delta = \frac{9}{2}\frac{(\theta - \sin\theta)^2}{(1 - \cos\theta)^3},$$

expand for small θ to find $\delta_{\rm lin} = \frac{3}{20}\left[6(\theta - \sin\theta)\right]^{2/3}$, and verify the numbers 5.55, 1.06 and 1.686 quoted in the text.